P510/2
PHYSICS
Paper 2
(Practical)
Sept. /Oct, 2026
3 hours

SET 3

Uganda Advanced Certificate of Education

PHYSICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two compulsory items.

Use **blue** *or* **black** *ink. Any work done in pencil* **except** *graphs and diagrams will* **not** *be scored.*

Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.

Details on the question paper should not be repeated in the responses.

Graph paper is provided.

Silent non-programmable scientific calculators may be used

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Item 1.

A school intends to provide magnifying glasses for learners to use when observing small objects during practical lessons. The school has run out of magnifying glasses with powers in the range **+4.0 D to +6.0 D** inclusive. However, in the laboratory there are several convex lenses whose specifications have faded and can no longer be read. One of the lenses has been selected and given to you by the laboratory technician. There is no one at the school who can investigate the specifications of the lens.

Task:

Plan and carry out a scientific investigation to determine the suitability of the lens provided to you.

Item 2.

Uganda Electricity Distribution Company Limited (UEDCL) uses electrical wires of specified resistance per metre length when installing and maintaining electricity distribution systems. A particular type of wire used for electricity distribution has a resistance per metre length in the range **2.10 – 8.20 Ω m^{-1}.** Recently, a section of the wire supplying electricity to a trading centre was damaged.

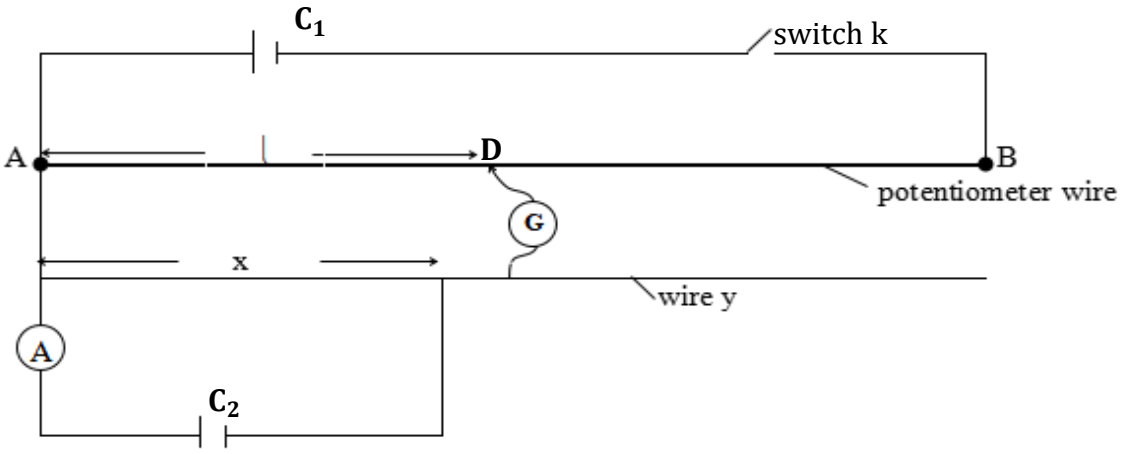
A UEDCL technician wishes to replace the damaged section. Available is a thin bare wire which the technician intends to use for the replacement. The technician remembers that one of the conditions prescribed in the manufacturer's specifications is that the replacement wire should have similar electrical properties to those of the original wire. However, the resistance per metre length of the wire the technician has is not known.

You are provided with the bare wire and the other apparatus required for the investigation.

Task:

Follow the instructions provided and use the result obtained to give feedback to the technician.

METHOD 1



(a) Connect the circuit as shown below

(b) Starting with $x = 10cm$, determine the position D on the potentiometer wire for which the galvanometer shows no deflection.

(c) Measure and record the balance length l and the reading, I, of the ammeter.

(d) Repeat procedures in (b) and (c) for $x = 30cm$

(e) Calculate the mean value of $\emptyset$

$$\emptyset = \frac{Ix}{l}$$

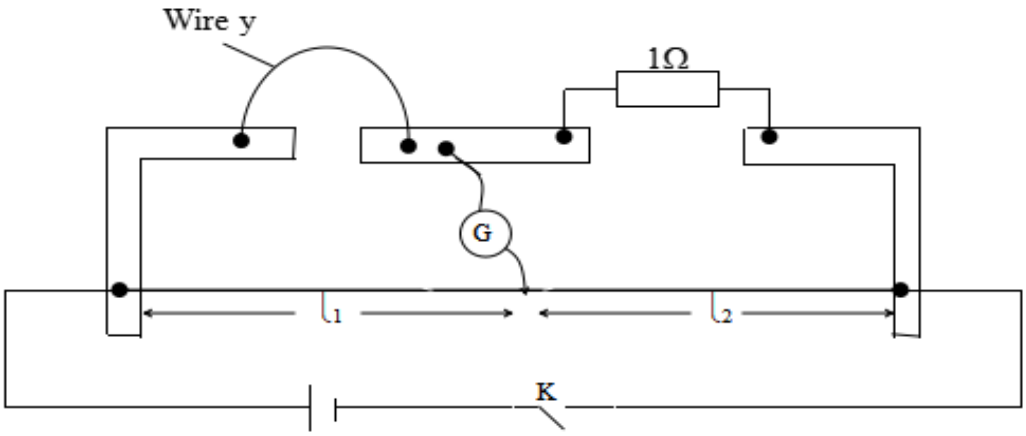
(f) Find the resistance per unit length, $\mathbf{k_1}$ of the wire from the equation

$$\mathbf{k_1} = \frac{1.5}{\emptyset}$$

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METHOD 2

a) Connect the circuit as shown below



b) Starting with a length $x = 0.10m$ of wire y, close switch K

c) Move the sliding contact along the potentiometer wire until a point is reached where the galvanometer shows no deflection.

d) Record the length l_1 and l_2

e) Open switch K

f) Repeat procedures c) to e) for values of x =0.150, 0.200, 0.250, 0.300, 0.350 and 0.400m.

(g) Tabulate your results including values of $\frac{l_1}{l_2}$ and $V = x(1 - x)$,

(h) Plot a graph of V against $\frac{l_1}{l_2}$

(i) Find the slope, S, of your graph.

(j) Calculate the resistance per metre, $\mathbf{k_2}$, of wire y from the expression $\mathbf{k_2 = \frac{1}{s}}$

End

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